Tracking Latent Reasoning Trajectories Using Rao-Blackwellized Particle Filtering

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Abstract

Monitoring the internal state of Large Language Models (LLMs) during complex reasoning is important for addressing "overthinking" and improving inference efficiency. This project treats reasoning trajectories as a Switching Linear Dynamic System (SLDS), where latent progress is obscured by noisy observations and potential hallucinations. We implement and compare three estimation methods: an Extended Kalman Filter (EKF), a standard Particle Filter (PF), and a Rao-Blackwellized Particle Filter (RBPF). Evaluated on a simulation grounded in the GSM8K dataset, results show that the RBPF outperforms unimodal estimators ($AUC \approx 0.71$ vs. $AUC \approx 0.13$ for EKF) by explicitly tracking discrete mode transitions. The findings show that analytical marginalization in the RBPF offers a sample-efficient solution for early success prediction in symbolic reasoning tasks.

1 Introduction

As Large Language Models (LLMs) perform increasingly complex multi-step reasoning, monitoring their internal "thinking" trajectories has become vital for AI safety and efficiency. While these models generate text, the underlying progress toward a solution is a hidden latent variable. While prior research indicates that cognitive states can be quantified via state-space models, mathematical reasoning often involves non-linear, discrete transitions that challenge traditional continuous estimators.

This project studies how to track these trajectories by modeling the reasoning process as a Switching Linear Dynamic System (SLDS). Treating reasoning steps as a noisy sensor feed, we developed a Rao-Blackwellized Particle Filter (RBPF) to analytically marginalize continuous progress while explicitly sampling for discrete mode switches. The model was validated using a simulation grounded in the step lengths of the GSM8K dataset. By benchmarking against Extended Kalman Filter (EKF) and standard Particle Filter baselines, this work uses metrics such as AUC to compare the performance of different estimators and show that methods based on marginalization perform better in this setting.

The primary contributions of this work are as follows:

- **Modeling of Reasoning as an SLDS:** We mapped symbolic logic trajectories to a 5-dimensional state space that accounts for both continuous drift and discrete switching dynamics.
- **Development of a Rao-Blackwellized Estimator:** We implemented a RBPF that analytically marginalizes continuous progress variables, improving sample efficiency and tracking accuracy during non-Gaussian transitions.

2 Related Work

The estimation of internal states from external observations has evolved from classical control theory to modern cognitive modeling. Early work by Yamauchi [1] established the feasibility of this approach by using a standard Kalman Filter to extract human "mindsets" from cursor motion trajectories. By inspecting system-level error-covariance, Yamauchi [1] showed that hidden mental states can be inferred from patterns in physical movements such as cursor trajectories. However, the reliance on linear models in that work limits the ability to handle the abrupt "jumps" or insights common in complex problem-solving.

In order to bridge the gap between physical systems and natural language, Miyoshi et al. [2] introduced the Language-Aided Particle Filter (LAPF), which showed that particle filters are uniquely suited for linguistic data due to their ability to represent high-dimensional observations as a probability distribution rather than a single point estimate. Recent surveys on latent reasoning [3] also suggest that these internal states can be modeled as continuous trajectories within a high-dimensional space. This is supported by work characterizing reasoning as a "chain of superposition" where hidden progress evolves through a sequence of overlapping latent states [4]. Furthermore, the paradigm of "Chain of Continuous Thought" (COCONUT) shows that LLMs can perform multi-step inference entirely within a continuous latent space rather than relying on language tokens [5].

In the specific context of Large Language Models (LLMs), Zhang [6] proposed the Confidence-Aware Reasoning (CaR) model to optimize performance by treating reasoning blocks as sequential decisions. While CaR focuses on pruning computation, this work extends these ideas by using an SLDS model. This allows for tracking of discrete mode transitions, such as "insights" or "backtracks", which are often modeled in other domains as Markov-switching processes [7]. To handle the resulting complexity, we use Rao-Blackwellized sampling, a technique established by Doucet et al. [8] to reduce estimator variance by analytically marginalizing the linear components of the model.

3 Method

This section describes the mathematical model used to track reasoning trajectories. We model the problem as a SLDS, where the latent state evolves according to both continuous drift and discrete mode transitions.

3.1 State Representation and System Dynamics

The reasoning state at time k is represented by a 5-dimensional vector $\mathbf{x}_k \in \mathbb{R}^5$. The components $\mathbf{x}_k = [p, c, u, f, m]^T$ encode progress (p), internal consistency (c), uncertainty (u), fatigue (f), and momentum (m). The state evolves according to a nonlinear stochastic update rule:

$$\mathbf{x}_{k+1} = R(\theta)(\mathbf{x}_k + \mathbf{f}_{drift}(\mathbf{x}_k)) + \mathbf{b}_z + \mathbf{w}_k, \quad \mathbf{w}_k \sim \mathcal{N}(0, Q)$$

where $z_{k+1} \in \{1, 2, 3\}$ is a discrete variable representing the system mode. Mode 1 represents steady-state progression, Mode 2 represents a high-gain state transition (impulse dynamics), and Mode 3 represents trajectory backtracking. The transition between modes is governed by a Markov chain with transition matrix P:

$$P = \begin{bmatrix} P_{11} & P_{12} & P_{13} \\ P_{21} & P_{22} & P_{23} \\ P_{31} & P_{32} & P_{33} \end{bmatrix}$$

where $P_{ij} = P(z_{k+1} = j | z_k = i)$. The state evolution incorporates a mode-dependent rotation $R(\theta)$ acting on the coherence-uncertainty subspace (c, u). For $z_{k+1} \in \{2, 3\}$, the rotation matrix is defined as:

$$R(\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) & 0 & 0 \\ 0 & \sin(\theta) & \cos(\theta) & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

with $\theta = 0.30$ rad for an INSIGHT mode and $\theta = -0.30$ rad for BACKTRACK. To ground the simulation in realistic reasoning scales, trajectory lengths T are dynamically mapped to the step-counts of the GSM8K dataset, to ensure the trajectory length reflects actual problem complexity.

3.2 Observation Model

The latent state is partially observable through a 3-dimensional measurement vector $\mathbf{y}_k$:

$$\mathbf{y}_k = h(\mathbf{x}_k) + \mathbf{v}_k, \quad \mathbf{v}_k \sim \mathcal{N}(0, R)$$

Where $h(\mathbf{x}_k)$ maps the hidden state to observable indicators such as coherence and uncertainty. To handle potential hallucinations in the reasoning trace, the model incorporates an observation likelihood with an outlier probability ϵ. When an outlier is detected, the measurement is assumed to originate from a distribution with an inflated covariance $R_{out} = \sigma_{scale} R$.

The progress variable p is excluded from the measurement function $h(\mathbf{x})$, which forces the estimators to infer latent success solely through the cross-correlations and rotational shifts observed in the remaining state components.

3.3 Extended Kalman Filter (EKF) Baseline

The EKF handles nonlinearities in $f(\cdot)$ and $h(\cdot)$ through local linearization. Because the reasoning dynamics are non-analytical, we use a numerical Jacobian J to approximate partial derivatives:

$$J_{ij} \approx \frac{f(\mathbf{x} + \epsilon \mathbf{e}_i)_j - f(\mathbf{x})_j}{\epsilon}$$

The EKF maintains a single Gaussian belief $\mathcal{N}(\boldsymbol{\mu}_k, \Sigma_k)$. However, since the EKF lacks a mechanism to represent the discrete mode z_k, it is forced to approximate switching impulses as continuous process noise, often leading to divergence during sudden high-gain transitions.

3.4 Standard Particle Filter (PF) Baseline

As an intermediate baseline, we implemented a standard Sampling Importance Resampling (SIR) Particle Filter. Unlike the RBPF, the standard PF does not use analytical marginalization. Instead, it represents the entire 6-dimensional joint state $(\mathbf{x}_k, z_k)$ through a set of discrete weighted particles $\{\mathbf{x}_k^{(i)}, z_k^{(i)}, w_k^{(i)}\}_{i=1}^N$.

Prediction: For each particle, the discrete mode is sampled from the Markov transition matrix P, and the continuous state is evolved through the nonlinear dynamics $f(\cdot)$ using the sampled mode's impulse.

Update and Resampling: Particles are weighted according to the observation likelihood $p(\mathbf{y}_k|\mathbf{x}_k^{(i)})$. To prevent particle deprivation, a common failure mode in high-dimensional estimation, we use systematic resampling whenever the effective sample size N_{eff} falls below a threshold $N/2$. This baseline allows us to isolate whether the performance gains in this project stem from the switching model itself or specifically from the Rao-Blackwellized marginalization.

3.5 Rao-Blackwellized Particle Filter (RBPF)

To overcome the limitations of unimodal estimation, we implement an RBPF. This filter uses the SLDS model by partitioning the state into a discrete mode

z_k (sampled via particles) and a continuous state $\mathbf{x}_k$ (tracked analytically).

By analytically marginalizing the continuous variables, which means integrating out the linear Gaussian components of the state using Kalman updates instead of sampling them, the RBPF reduces the variance of the importance weights compared to the standard PF described in the previous section.

Prediction and Analytical Tracking: For each particle i, a new mode $z_{k+1}^{(i)}$ is sampled from $P(z_{k+1}|z_k^{(i)})$. Conditioned on this sampled mode, the continuous belief $\mathcal{N}(\boldsymbol{\mu}_k^{(i)}, \Sigma_k^{(i)})$ is updated analytically using the EKF equations. This allows each particle to maintain its own unique Gaussian trajectory adapted to its specific mode history.

Likelihood Weighting: Instead of weighting by a point estimate, particles are weighted by the marginal likelihood of the observation $\mathbf{y}_k$. We use a mixture model to account for potential hallucinations:

$$w_k^{(i)} \propto w_{k-1}^{(i)} \cdot \left[(1-\epsilon)\mathcal{N}(\mathbf{y}_k; \hat{\mathbf{y}}_k^{(i)}, S_k^{(i)}) + \epsilon\mathcal{N}(\mathbf{y}_k; \hat{\mathbf{y}}_k^{(i)}, \sigma_{scale} S_k^{(i)})\right]$$

where $\hat{\mathbf{y}}_k^{(i)}$ and $S_k^{(i)}$ are the mode-conditioned predicted measurement and innovation covariance. This partitioning allows the filter to maintain a multimodal posterior, which is important for capturing the discrete state transitions identified in the experiments.

4 Experiments

We evaluated the tracking performance of the RBPF against the EKF and a standard PF using the GSM8K-grounded simulation. To isolate the impact of latent dynamics, all trajectories are initialized at the same state $\mathbf{x}_0$, to eliminate bias from differing initial conditions.

4.1 Tracking Dynamics and Mode Detection

The qualitative behavior of the filters during a single successful reasoning trace is shown in 2. While the raw sensor signals in 2A are highly stochastic, the RBPF identifies the discrete state transition at Step 3. This is visible in the posterior mode probability in 2B, where the belief in a state jump transitions from 0 to 1.

As seen in 2C, the EKF diverges downward because its unimodal Gaussian constraint forces it to interpret noisy steps as evidence of failure. By contrast, the RBPF correctly revises the progress estimate upward once the mode switch is detected.

4.2 Statistical Performance

The aggregate discriminative power is summarized by the ROC curves in 1. The RBPF achieves higher performance ($AUC \approx 0.71$) over the standard PF ($AUC \approx 0.57$). The EKF fails to show early predictive power, acting as an inverse classifier in this regime ($AUC \approx 0.13$).

The shaded regions in 1 illustrate variability across trials, which is especially high for the standard PF, while the RBPF remains more stable. This suggests that prediction sensitivity depends on the stochastic timing of the discrete state transition.

4.3 Computational Scaling

Finally, 3 shows how accuracy scales with the particle count N. Both the RBPF and standard PF saturate around $N = 50$, suggesting they are relatively sample-efficient for this state space. However, even at high particle counts, the EKF remains at a near-zero baseline. This shows that the failure of Gaussian assumptions is due to a mismatch between the model and the underlying system, and cannot be solved by simply increasing computational resources.

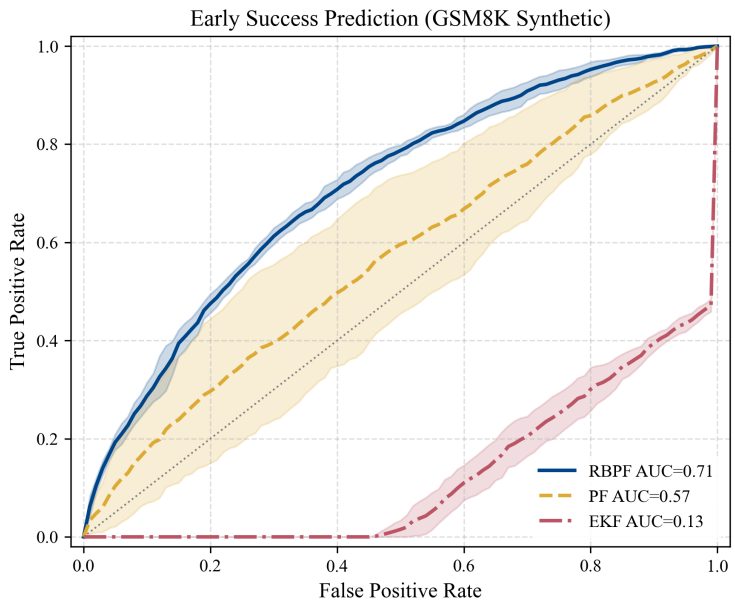

Figure 1: ROC curves for early success prediction. The RBPF (AUC ≈ 0.71) leads the standard PF (AUC ≈ 0.57), while the EKF ($AUC \approx 0.13$) fails. Shaded bands indicate that performance is sensitive to the specific timing of discrete state transitions across trials.

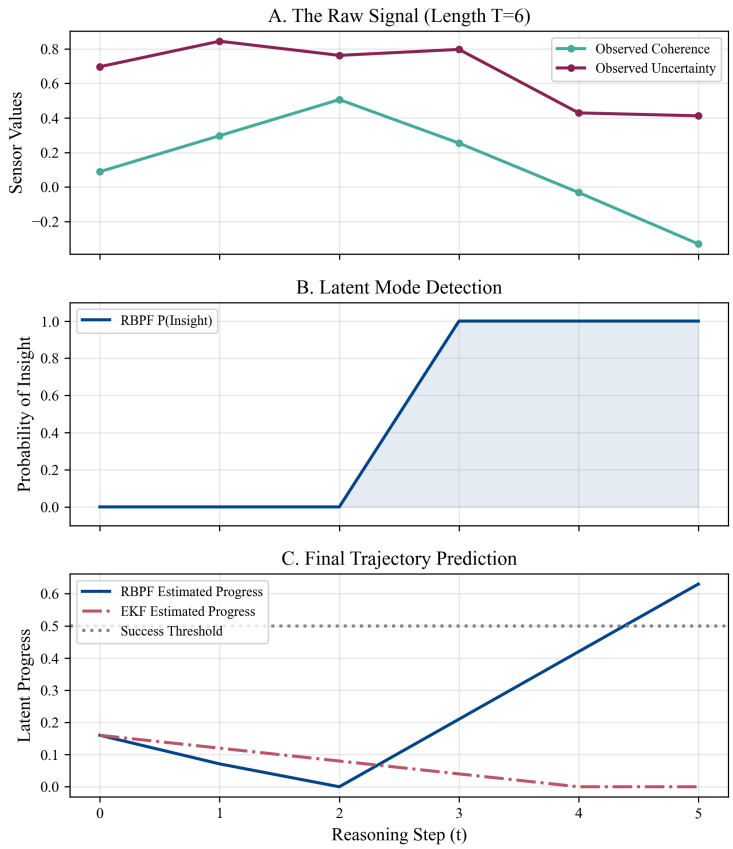

Figure 2: Single-trajectory tracking. (A) Noisy sensor signals. (B) RBPF detection of a mode transition at Step 3. (C) RBPF upward revision of the progress estimate compared to EKF divergence.

5 Conclusion

5.1 Summary of Findings

This study examined the application of Bayesian state estimation to the problem of monitoring latent progress in mathematical reasoning. By modeling the reasoning process as an SLDS, we showed that traditional unimodal estimators, such as the EKF, are fundamentally limited by Gaussian assumptions. The EKF demonstrated an inability to track discrete mode transitions, resulting in estimator divergence ($AUC \approx 0.13$). Conversely, the RBPF captured these non-linear transitions, achieving the highest performance ($AUC \approx 0.71$) while remaining computationally efficient at low particle counts.

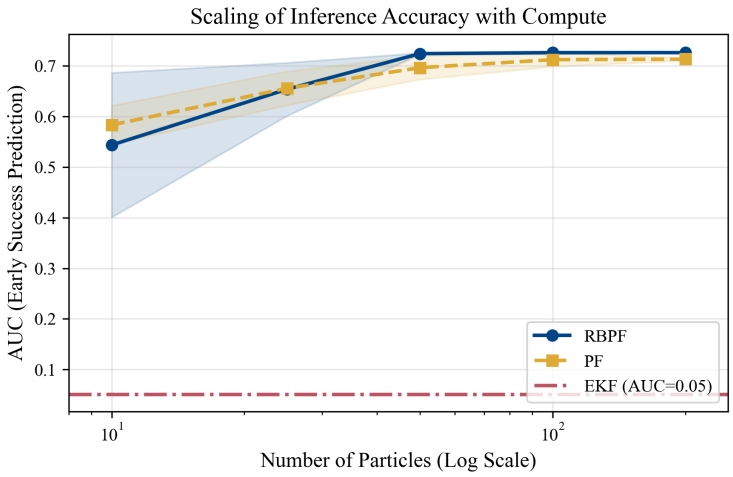

Figure 3: Accuracy versus particle count. Both sampling filters saturate at 50–100 particles. The horizontal EKF baseline highlights that increasing compute cannot overcome incorrect Gaussian assumptions in switching systems.

5.2 Analysis of Estimator Behavior

The primary insight from this project is that reasoning is not a continuous drift, but a series of high-gain updates or mode switches. The EKF failure seen in 2C is not a matter of noise, but a model mismatch. Because the EKF assumes a single mean, it is forced to compromise between a successful and a failed trajectory when noise is high, leading to the downward bias observed. The RBPF succeeds because of its ability to represent multiple hypotheses of reasoning paths, waiting for evidence of a mode switch without discarding the possibility of success. This property is important for real-world applications where stochastic fluctuations in early reasoning steps might otherwise trigger premature stopping.

5.3 Future Work and Limitations

While the RBPF performs well, it is still sensitive to the specific parameters of the transition matrix and the timing of state jumps. A limitation of the current model is the fixed nature of the observation noise. In actual reasoning trajectories, noise levels often fluctuate depending on the complexity of the task. Future research could study learning models that operate directly in continuous latent spaces to eliminate the need for hand-crafted transition dynamics. Additionally, integrating token-level log-probabilities as a dynamic sensor feed could allow the filter to adaptively tune its gain,

potentially further reducing the variance seen in early predictions.

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